Major Project

# Work Hub– Multi-Tenant Enterprise Management Platform

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## Abstract

Modern organizations require efficient software systems to manage employee information, attendance, payroll, project activities, and internal communication. However, many small and medium-sized organizations still rely on manual processes or disconnected software tools, resulting in data inconsistency, operational inefficiency, and delayed decision-making. Traditional monolithic enterprise applications further increase maintenance complexity and limit scalability due to tightly coupled system components.

This project presents the design and development of a microservices-based enterprise management system using Angular for the frontend and Spring Boot for the backend. The system integrates employee management, session-based automatic attendance tracking, leave management, payroll processing, project and task management, performance analytics, and real-time notifications. By adopting a modular microservices architecture, the proposed system improves scalability, maintainability, and real-time responsiveness, making it suitable for academic implementation as well as real-world organizational use.


## Introduction

Efficient management of organizational operations such as employee records, attendance tracking, payroll processing, and project coordination is essential for improving productivity and ensuring smooth business functioning. However, many organizations, particularly small and medium-sized enterprises, rely on manual processes or multiple disconnected applications to handle these activities.

Traditional enterprise management systems are typically built using monolithic architectures, where all functionalities are tightly integrated into a single application. As system complexity grows, such architectures become difficult to maintain and scale. Microservices architecture addresses these challenges by dividing the system into independent services, each responsible for a specific business function.

This project focuses on developing a microservices-based enterprise management system that automates core organizational processes while maintaining simplicity and usability. The system aims to reduce manual effort, improve operational efficiency, and enable real-time organizational communication.

## Problem Statement

To design and develop a scalable enterprise management system that automates employee operations such as attendance, payroll, project tracking, and notifications using a microservices architecture.

## Existing System

In many organizations, employee operations are managed using manual processes or isolated software tools. Attendance is recorded manually, payroll is calculated using spreadsheets, and project tracking relies on informal communication methods such as emails or messaging applications.

## Drawbacks of the Existing System

- Manual processes are time-consuming and error-prone
- Lack of integration between organizational functions
- No real-time notifications or updates
- Poor visibility into employee performance
- Difficult to scale as organization grows

## Proposed System

The proposed system is a web-based enterprise management platform developed using microservices architecture. It automates employee management, attendance tracking, payroll processing, project and task monitoring, performance analytics, and real-time notifications while ensuring scalability and maintainability.

## System Modules and Their Functionalities

### Organization Management Module

- Organization and department management
- Role and designation handling
- Organization-level configuration

### Employee Management Module

- Employee profile management
- Role and department assignment
- Employee status tracking

### Attendance and Leave Management Module

- Automatic attendance based on login/logout
- Leave request and approval workflow
- Leave balance tracking

### Payroll Management Module

- Salary configuration
- Attendance-based payroll calculation
- Payroll reports and salary history

### Project and Task Management Module

- Project creation and assignment
- Task tracking and deadline monitoring

### Performance Analytics Module

- Task completion analysis
- Employee productivity reports
- Dashboard visualization

### Notification Module

- Real-time system notifications
- Leave and payroll alerts
- WebSocket-based updates

## Backend Microservices Architecture

### Authentication and Organization Microservice

- Handles user authentication and authorization using token-based security
- Manages roles, permissions, organizations, and departments

- Provides identity validation services to other microservices
- Operates independently; failure affects only login and authorization, while already authenticated sessions in other services remain functional

### Employee, Attendance and Leave Microservice

- Manages employee records and employment status
- Automatically tracks attendance based on login and logout sessions
- Handles leave application, approval, and leave balance calculation
- Functions independently; failure impacts only employee, attendance, and leave operations without affecting payroll, project tracking, or authentication services

### Payroll, Project and Performance Microservice

- Calculates payroll based on attendance and salary configuration
- Manages projects, tasks, priorities, and deadlines
- Generates basic performance analytics and reports
- Operates independently; failure affects payroll and task-related features while employeemanagement and attendance services continue to function normally

### Notification Microservice

- Handles real-time notifications
- Sends alerts for attendance, leave approvals, task assignments, and payroll generation
- Processes system events asynchronously
- Fully independent and non-blocking; failure does not affect core business logic or dataprocessing in other microservices

## Technology Stack

Frontend: Angular, HTML5, CSS3, TypeScript

Backend: Spring Boot, REST APIs, JWT

Database: MySQL

Tools: Git, GitHub, Postman, IntelliJ, VS Code

## Hardware Requirements

- Processor: Intel i5 or equivalent
- RAM: Minimum 8 GB
- Storage: 256 GB SSD or higher

## Conclusion

The proposed microservices-based enterprise management system provides a scalable and efficient solution for managing organizational operations. By automating attendance, payroll, project tracking, and notifications, the system reduces manual effort and improves productivity. The modular microservices architecture ensures maintainability, scalability, and future extensibility, making the project suitable for both academic evaluation and real-world deployment.